

LAB OPERATIONS SUPPLY CHAIN ANALYTICS

End-to-End Microsoft Fabric and Power BI Portfolio Project

An analytics solution that transforms raw laboratory purchasing, inventory and material-usage data into clean dimensional models and interactive business intelligence reports.

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Data and Operations Analytics Portfolio

August 2026

Microsoft Fabric • Data Factory • Dataflow Gen2 • Lakehouse • Warehouse • SQL • Power BI • DAX • Git

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Business Problem and Project Objectives

Business Problem

Laboratory purchasing, inventory-count and material-usage data was distributed across five separate CSV files. The data contained exact duplicates, missing values, inconsistent date formats and invalid reference records. Without a unified analytics model, stakeholders lacked a reliable way to evaluate purchasing spend, vendor performance, inventory accuracy and material consumption.

Project Objectives

- Consolidate the five source datasets within Microsoft Fabric.
- Apply Bronze, Silver and Gold architecture to separate raw, cleaned and analytics-ready data.
- Profile, standardize and deduplicate the source data.
- Build a dimensional Warehouse model using shared dimensions and separate fact tables.
- Create reusable DAX measures and interactive Power BI reports for purchasing, inventory and usage analysis.
- Automate Bronze-to-Silver processing through a Fabric pipeline and Dataflow Gen2.

Solution Architecture and Data Flow

Raw CSVs → Bronze Lakehouse → Dataflow Gen2



Silver Lakehouse → Warehouse Staging → Gold Star Schema



Semantic Model → Power BI Reports

Bronze and Silver Lakehouse

Explorer

Add lakehouses

SupplyChainLakehouse

- Tables
- Files
 - Bronze

«

»

Search

Bronze

X

FabricProject > SupplyChainLakehouse > Files > Bronze

Name
inventory_counts.csv
lots.csv
materials.csv
outtakes.csv
purchases.csv

Five raw source CSV files stored in the Bronze folder of the SupplyChain Lakehouse.

SupplyChainLakehouse

- Tables
 - dbo
- Bronze
 - inventory_counts
 - lots
 - materials
 - outtakes
 - purchases
- Silver
 - inventory_counts
 - lots
 - materials
 - outtakes
 - purchases

Bronze tables preserve the source data, while Silver tables contain standardized and deduplicated records for downstream analytics.

Data Transformation and Automation

Queries [5]

Bronze inventory_counts

Bronze materials

Bronze outtakes

Bronze purchases

Bronze lots (2)

Bronze inventory...

ABC123

6 steps

Table.Distinct("#Changed column type", {"count_line_id"})

	count_line_id	count_event_id	count_date	facility_id	storage_location	material_id	lot_id	system_quar
1	CNTL-0000001	CNTE-20260630-OAK	6/30/2026	FAC-OAK	OAK-STOR-A1	MAT-0001	LOT-000002	
2	CNTL-0000002	CNTE-20250630-SSF	6/30/2025	FAC-SSF	SSF-STOR-A1	MAT-0001	LOT-000003	
3	CNTL-0000003	CNTE-20250930-SSF	9/30/2025	FAC-SSF	SSF-STOR-A1	MAT-0001	LOT-000003	
4	CNTL-0000004	CNTE-20251231-SSF	12/31/2025	FAC-SSF	SSF-STOR-A1	MAT-0001	LOT-000003	
5	CNTL-0000005	CNTE-20260331-SSF	3/31/2026	FAC-SSF	SSF-STOR-A1	MAT-0001	LOT-000003	
6	CNTL-0000006	CNTE-20251231-OAK	12/31/2025	FAC-OAK	SSF-STOR-C3	MAT-0001	LOT-000004	
7	CNTL-0000007	CNTE-20260331-OAK	3/31/2026	FAC-OAK	SSF-STOR-C3	MAT-0001	LOT-000004	
8	CNTL-0000008	CNTE-20250630-SSF	6/30/2025	FAC-SSF	OAK-STOR-B2	MAT-0001	LOT-000005	
9	CNTL-0000009	CNTE-20251231-SSF	12/31/2025	FAC-SSF	OAK-STOR-B2	MAT-0001	LOT-000005	
10	CNTL-0000010	CNTE-20260630-SSF	6/30/2026	FAC-SSF	OAK-STOR-B2	MAT-0001	LOT-000005	
11	CNTL-0000011	CNTE-20250331-SSF	3/31/2025	FAC-SSF	OAK-STOR-B2	MAT-0001	LOT-000006	

Query settings

Properties

Name
Bronze inventory_counts

Applied steps

Source

Navigation

Navigation 1

Navigation 2

ABC123 Changed column type

Removed duplicates

Data destination

Lakehouse

Dataflow Gen2 standardizes column types, removes duplicate business keys and writes cleaned records to the Silver Lakehouse.

Dataflow

Run_Bronze_to_Silver

Parameters

Variables

Settings

Output

Library variables

Pipeline run ID

391c1cab-26ec-49cf-a38c-574ad8618828

View run detail

Succeeded

Export to CSV

Filter

Column options

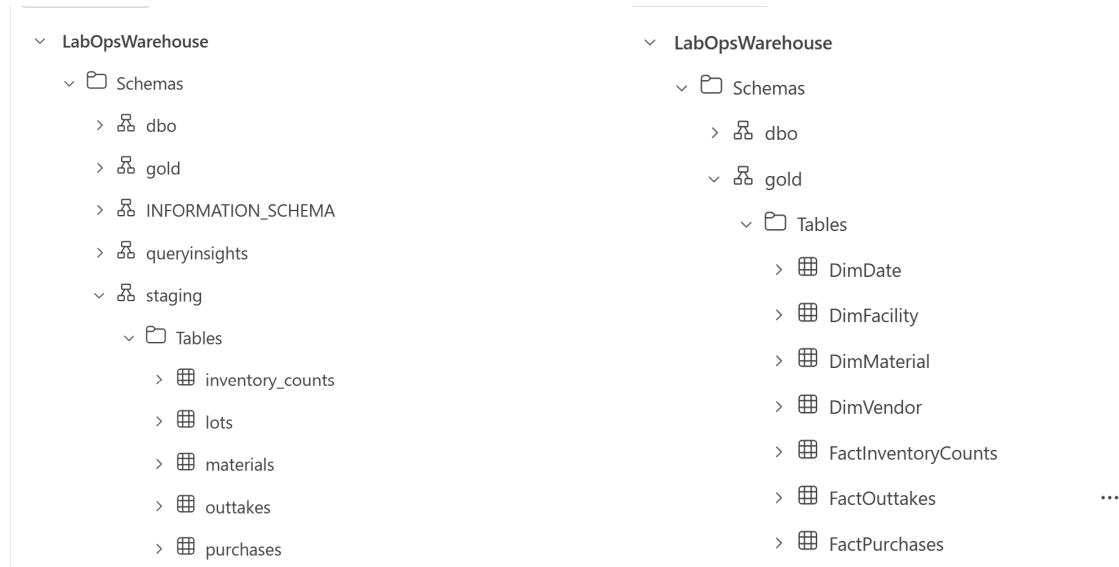
Filter by keyword

Showing 1 - 1 items

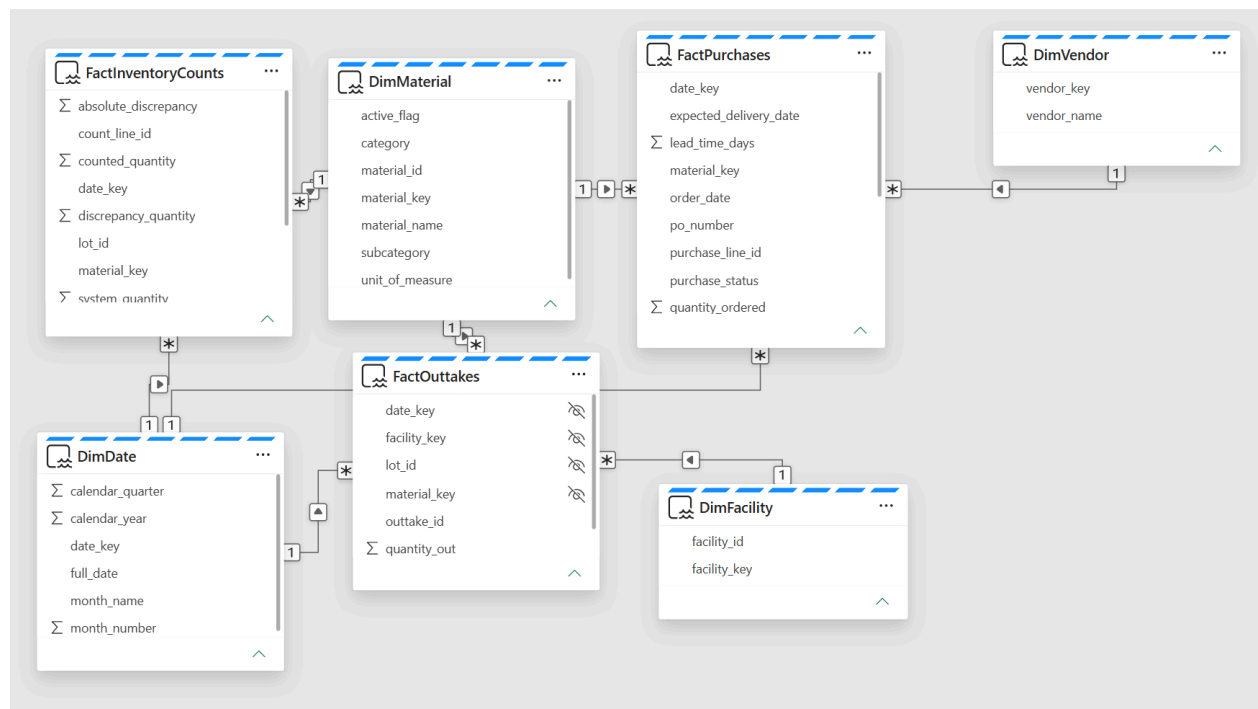
Activity name	Activity status	Run start	Duration	Input	Output
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The Fabric pipeline successfully executed the automated Bronze-to-Silver transformation workflow.

Warehouse and Semantic Model

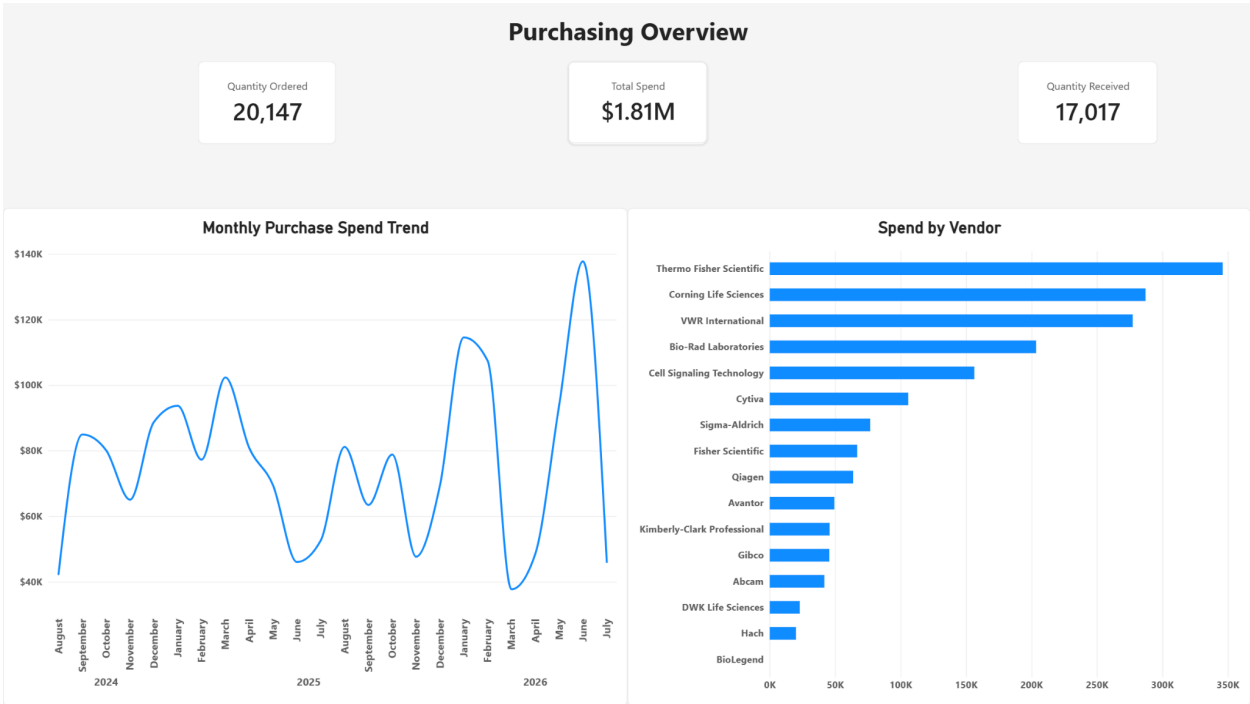


Cleaned Silver data is loaded into the Warehouse staging schema before being organized into analytics-ready Gold fact and dimension tables.

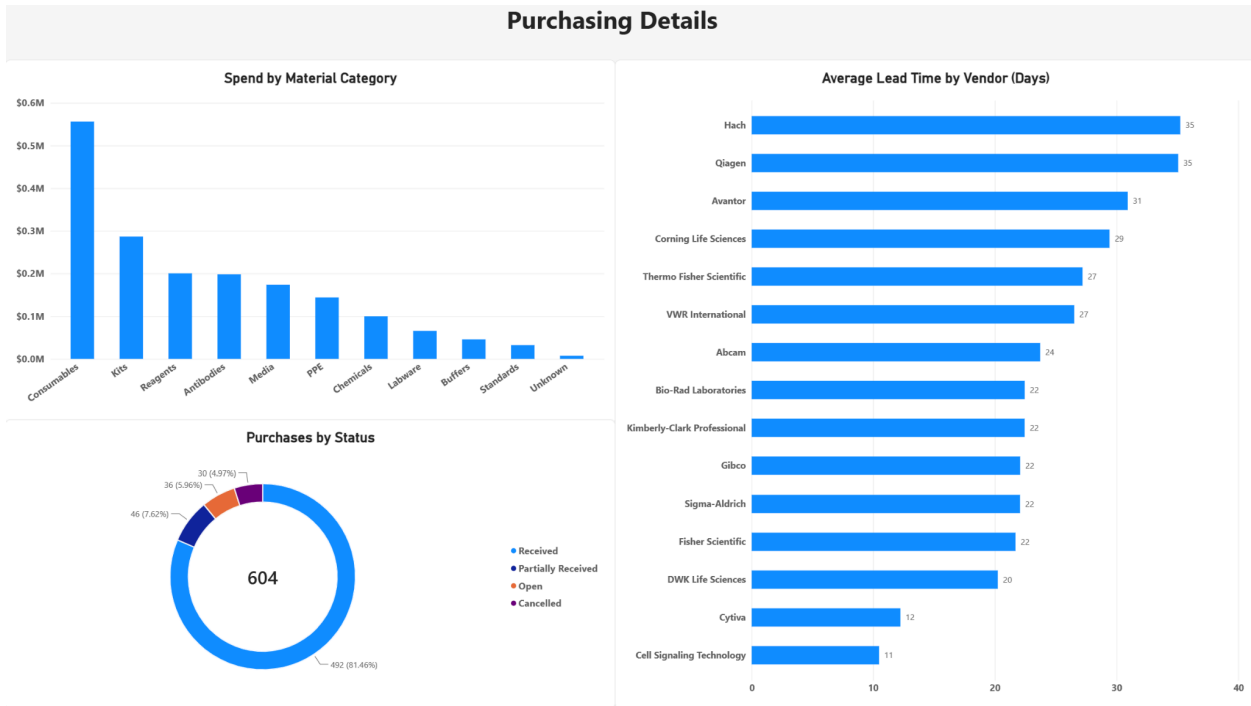


The semantic model connects shared dimensions to purchasing, inventory-count and material-usage fact tables through one-to-many relationships.

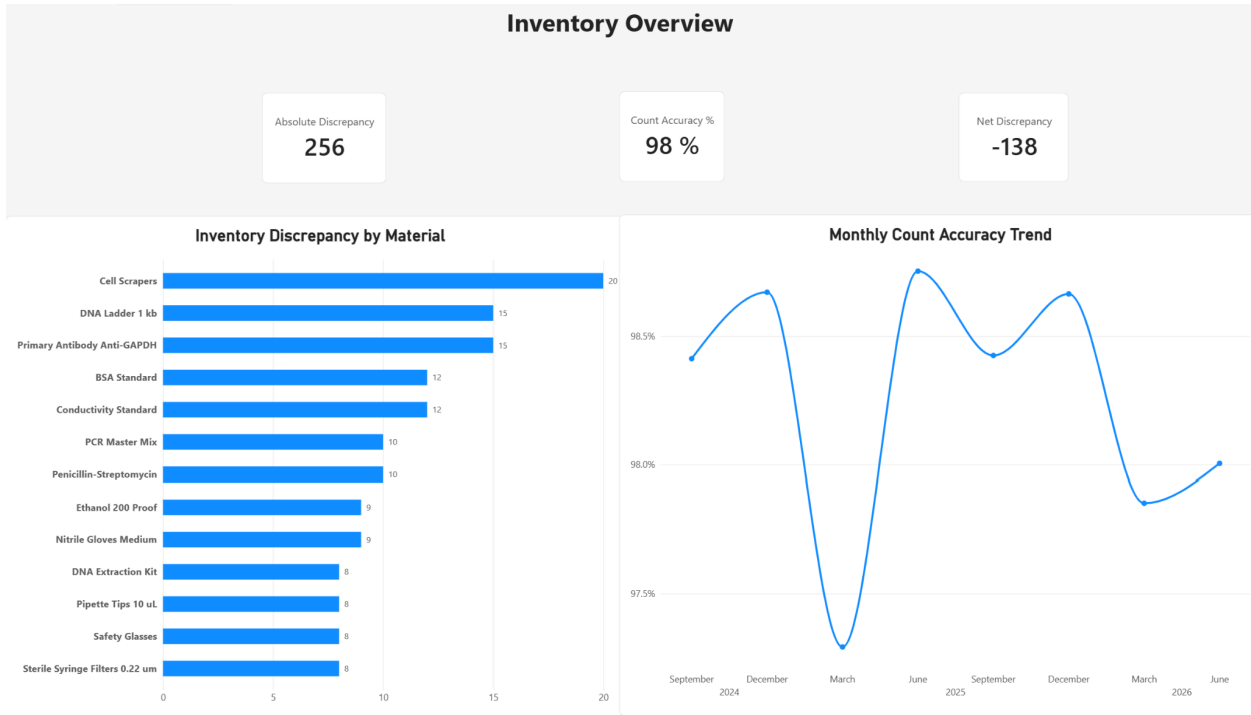
Purchasing Overview



Purchasing Details

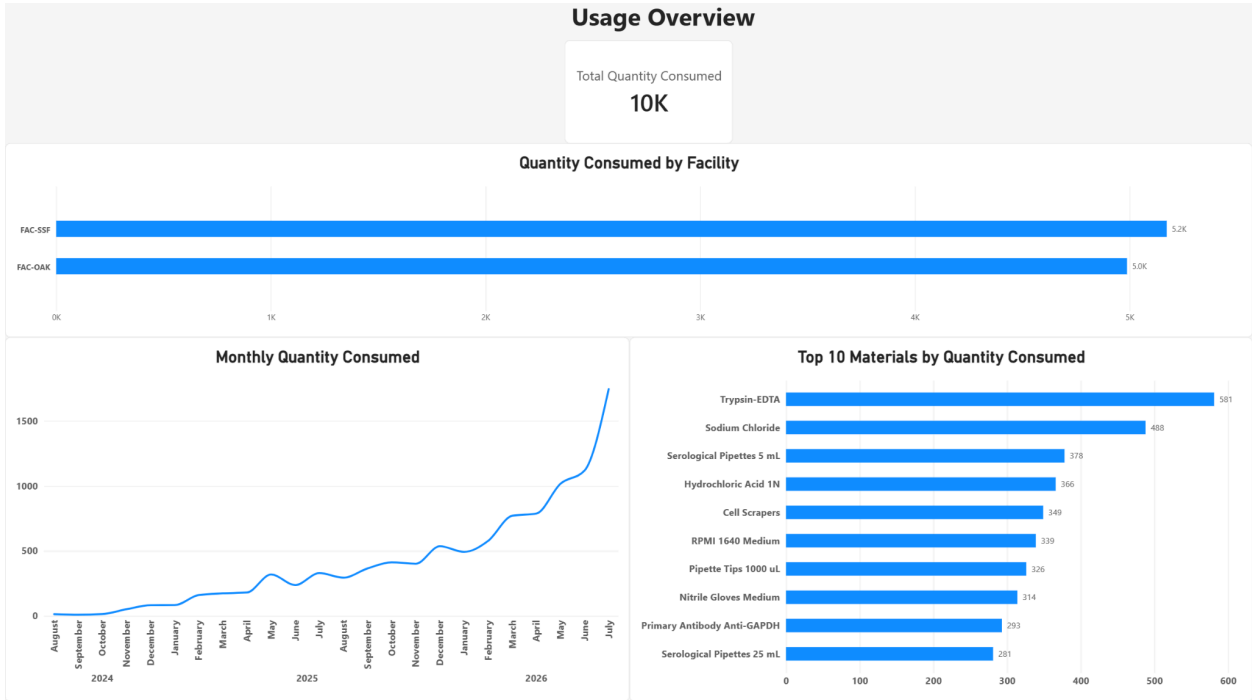


Inventory Overview



The Inventory Overview report tracks count accuracy, absolute and net discrepancies, and the materials contributing most heavily to inventory variance.

Usage Overview



The Usage Overview report summarizes total material consumption and highlights monthly trends, high-usage materials and consumption by facility.

Key Findings and Skills Demonstrated

Key Findings

- Purchasing activity totaled approximately \$1.81 million across the analyzed period.
- Inventory counts achieved 98% overall accuracy.
- Absolute inventory discrepancies totaled 256 units, while the net discrepancy was –138 units.
- Approximately 10,000 units of material were consumed.
- Vendor, material and facility-level analysis identified the primary drivers of purchasing spend, inventory variance and material usage.

Skills Demonstrated

- Microsoft Fabric medallion architecture
- Data profiling, cleansing and deduplication
- Dataflow Gen2 and pipeline orchestration
- Lakehouse and Delta tables
- SQL Warehouse development
- Dimensional and semantic modeling
- DAX measures and Power BI reporting
- Git-based version control